

Notes on Probabilistic and Unsupervised Learning

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Fall 2024

1 Fundamental Learning Problems

In machine learning, we are often concerned with learning from data. We can broadly categorize the learning problems into three types.

Definition 1.1 (Supervised Learning). In **supervised learning**, we are given a dataset of input-output pairs, often called a "training" set:

$$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

The goal is to learn a function f that maps inputs to outputs, $f : X \rightarrow Y$, such that we can predict the correct output y^* for a new, unseen input x^* . The provided outputs y_i act as the "supervision" or "teaching" signal.

Remark 1.2 (Classification vs. Regression). Supervised learning (Definition 1.1) problems can be further divided into two main categories:

1. **Classification:** The output values y_i belong to a finite set of discrete categories or "classes". The goal is to learn a decision boundary that separates the classes.
2. **Regression:** The output values y_i are continuous. The goal is to fit a function to the data that captures the underlying relationship between inputs and outputs.

Statistically speaking, both are forms of regression where we fit a function to data.

Definition 1.3 (Unsupervised Learning). In **unsupervised learning**, we are given a dataset consisting only of input data, without any corresponding output labels:

$$D = \{x_1, x_2, \dots, x_n\}$$

The goal is to discover interesting structures, patterns, or representations in the data. This could involve describing the data's probability distribution $p(x)$, identifying underlying structural variables, or grouping similar data points.

Example 1.4 (Classification vs. Clustering). Consider a dataset of points in $\mathbb{R}^2$ that belong to one of three groups.

- **Classification (Supervised):** We are given the data points *and* their group assignments (classes), as in Definition 1.1. The task is to learn a function that can assign a class to *new*, unseen points.
- **Clustering (Unsupervised):** We are given only the data points, as in Definition 1.3. The task is to assign each point *in the given dataset* to a group (a cluster) based on the structure of the data itself. The solution is typically a partitioning of the data, which can be represented by a distribution over cluster assignments.

Definition 1.5 (Reinforcement Learning). In **reinforcement learning**, an agent interacts with an environment over time. The agent observes a state x_t , takes an action a_t , and receives a reward (or payoff) r_t . The sequence of interactions can be represented as:

$$x_1 \xrightarrow{a_1} r_1, \quad x_2 \xrightarrow{a_2} r_2, \quad x_3 \xrightarrow{a_3} r_3, \dots$$

The goal is to learn a **policy**, which is a mapping from states to actions, that maximizes the cumulative payoff over time.

2 The Probabilistic Approach to Learning

Our approach to learning begins with a model of how data is produced. In machine learning, this is often called a **generative model**.

Definition 2.1 (Statistical Model). Let data $\mathcal{D} = (x_1, \dots, x_n)$ be observed from a data source, where each observation lies in a **sample space** $\mathcal{X}$.

1. Our mathematical description of the source is a probability distribution P on $\mathcal{X}$.
2. A **(statistical) model**, denoted $\mathcal{M}$, is a set of probability distributions. The assumption "I assume model $\mathcal{M}$ " means "I assume the true data-generating distribution P is contained in $\mathcal{M}$ ".
3. We typically index the elements of $\mathcal{M}$ by a **parameter** θ from a **parameter space** $\mathcal{T}$. The model then takes the form:

$$\mathcal{M} = \{P(\cdot|\theta) \mid \theta \in \mathcal{T}\} \tag{1}$$

Example 2.2 (Gaussian Location Model). Let the sample space be $\mathcal{X} = \mathbb{R}$. A simple model (Definition 2.1) could be the set of all Gaussian distributions with variance 1:

$$\mathcal{M} = \{\text{Gaussian}(\theta, 1) \mid \theta \in \mathbb{R}\}$$

Here, the parameter θ is the mean of the Gaussian, and the parameter space is $\mathcal{T} = \mathbb{R}$.

2.1 Basic Rules of Probability

Let x and y be random variables.

- Proposition 2.3** (Properties of Probability Distributions). 1. **Non-negativity:** For any event x , $P(x) \geq 0$.
2. **Normalization:** If x is discrete, $\sum_{x \in \mathcal{X}} P(x) = 1$. If x is continuous with probability density function $p(x)$, then $\int_{\mathcal{X}} p(x) dx = 1$.
3. **Joint Probability:** $P(x, y)$ is the probability of observing both x and y .
4. **Marginal Probability:** The probability of x is obtained by summing or integrating over all possible values of y : $P(x) = \sum_y P(x, y)$ (for discrete y).
5. **Conditional Probability:** The probability of x given y is $P(x|y) = \frac{P(x, y)}{P(y)}$, assuming $P(y) > 0$.

Theorem 2.4 (Bayes' Formula). *From the definition of conditional probability (Proposition 2.3), we have the chain rule: $P(x, y) = P(x|y)P(y) = P(y|x)P(x)$. Rearranging this gives Bayes' formula:*

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)} \quad (2)$$

In the context of statistical inference, this is often written as:

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}$$

Remark 2.5 (Notation Convention). We will often use P for probability mass functions (discrete) and p for probability density functions (continuous), but this distinction may be relaxed where the meaning is clear from context.

Definition 2.6 (Independence). Two random variables X_1 and X_2 with marginal distributions P_1 and P_2 are **independent** if their joint distribution P factorizes:

$$P(x_1, x_2) = P_1(x_1)P_2(x_2)$$

Equivalently, $P(x_1|x_2) = P_1(x_1)$. Informally, knowing the value of one variable provides no information about the other.

Definition 2.7 (IID Random Variables). A sequence of random variables $X_1, \dots, X_n$ is called **independent and identically distributed (i.i.d.)** if they are mutually independent (Definition 2.6) and all have the same marginal distribution, P . In this case, their joint distribution is:

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i) \quad (3)$$

3 Exponential Families

Definition 3.1 (Exponential Family). A model $\mathcal{M} = \{p(\cdot|\theta) \mid \theta \in \mathcal{T}\}$ is an **exponential family** if its probability density or mass function can be written in the form:

$$p(x|\theta) = f(x)g(\theta)e^{\phi(\theta)^T T(x)} \quad (4)$$

for all $x \in \mathcal{X}$ and $\theta \in \mathcal{T}$. The components are:

- $T(x)$: A function $T: \mathcal{X} \rightarrow \mathbb{R}^m$ called the **sufficient statistic**.
- $\phi(\theta)$: A function $\phi: \mathcal{T} \rightarrow \mathbb{R}^m$ whose values are called the **natural parameters**.
- $f(x)$: The base measure, a non-negative function.
- $g(\theta)$: A positive function that acts as a normalizer, ensuring the distribution integrates/sums to 1. It is fully determined by T , f , and ϕ .

Remark 3.2 (Ubiquity of Exponential Families). Exponential families are arguably the most common class of distributions in machine learning. They include most elementary distributions (Gaussian, Bernoulli, Beta, etc.) and are convenient to work with mathematically. They often serve as building blocks for more complex models.

Example 3.3 (Common Exponential Families). The following table shows some common distributions that are members of the exponential family (Definition 3.1) and their sufficient statistics $T(x)$.

Model	Sample Space	Sufficient Statistic $T(x)$
Gaussian	$\mathbb{R}^d$	(xx^T, x)
Gamma	$\mathbb{R}_+$	$(\ln(x), x)$
Poisson	$\mathbb{N}_0$	x
Bernoulli	$\{0, 1\}$	x
Beta	$[0, 1]$	$(\ln(x), \ln(1 - x))$
Dirichlet	Prob. dists. on d events	$(\ln x_1, \dots, \ln x_d)$

A non-example is the Student's t -distribution, as this distribution has heavy tails that do not decay exponentially.

4 Parameter Estimation

Given a model $\mathcal{M} = \{P(\cdot|\theta)\}$ (Definition 2.1) and data $\mathcal{D}$, the goal of learning is to select a "best" parameter θ . We distinguish two main philosophies for this task.

Definition 4.1 (Point Estimation). **Point estimation** assumes there is a single true, but unknown, parameter value θ_0 . The goal is to produce an estimate $\hat{\theta}$, which is a function of the observed data, that approximates θ_0 .

Definition 4.2 (Bayesian Inference). **Bayesian inference** treats the parameter θ itself as a random variable Θ . Our uncertainty about Θ is represented by a probability distribution.

- We must first specify a **prior distribution**, $p(\theta)$, which represents our beliefs about Θ before observing any data.
- After observing data $\mathcal{D}$, we update our beliefs using Bayes' rule (Theorem 2.4) to compute the **posterior distribution**, $p(\theta|\mathcal{D})$.

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})} \quad (5)$$

The result of the inference is the entire posterior distribution, not a single point estimate.

4.1 Specific Estimation Methods

Definition 4.3 (Maximum Likelihood Estimation). The **Maximum Likelihood Estimation (MLE)** is the most common form of point estimation (Definition 4.1). It chooses the parameter θ that maximizes the likelihood of observing the data $\mathcal{D}$.

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta} p(\mathcal{D}|\theta) \quad (6)$$

Here, $p(\mathcal{D}|\theta)$ is the **likelihood function**, often denoted $\mathcal{L}(\theta; \mathcal{D})$.

Definition 4.4 (Maximum a Posteriori Estimation). The **Maximum a Posteriori (MAP)** estimate is the value of θ that is most probable under the posterior distribution (Definition 4.2). It is the mode of the posterior.

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta|\mathcal{D}) = \arg \max_{\theta} p(\mathcal{D}|\theta)p(\theta) \quad (7)$$

The denominator $p(\mathcal{D})$ is dropped as it does not depend on θ .

Remark 4.5 (MAP as Regularized MLE). Although the MAP estimate (Definition 4.4) is defined in terms of a posterior distribution (a Bayesian concept), it is a point estimate. It can be viewed as a regularized version of the MLE (Definition 4.3), where the prior $p(\theta)$ penalizes certain parameter values.

5 Bayesian Learning in Detail

5.1 Modeling Assumptions and Posterior

Definition 5.1 (Bayesian Model). A standard Bayesian model consists of a pair $(\mathcal{M}, \pi)$, where $\mathcal{M} = \{p(\cdot|\theta) \mid \theta \in \mathcal{T}\}$ is a statistical model (Definition 2.1) and π is a prior distribution on the parameter space $\mathcal{T}$.

The data is assumed to be generated via the following process:

1. A parameter Θ is drawn from the prior: $\Theta \sim \pi$.
2. Given the parameter Θ , data points $X_1, \dots, X_n$ are drawn independently from the corresponding distribution: $X_1, \dots, X_n | \Theta \sim_{\text{iid}} p(\cdot|\Theta)$.

This means the data $X_1, \dots, X_n$ are **conditionally i.i.d.** (Definition 2.7) given Θ .

Proposition 5.2 (Posterior Distribution). *Under the conditional i.i.d. assumption, the likelihood of the data $\mathcal{D} = \{x_1, \dots, x_n\}$ is $p(\mathcal{D}|\theta) = \prod_{i=1}^n p(x_i|\theta)$. Bayes' rule (Theorem 2.4) then gives the posterior:*

$$\pi(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)\pi(\theta)}{\int_{\mathcal{T}} p(\mathcal{D}|\theta')\pi(\theta')d\theta'} = \frac{(\prod_{i=1}^n p(x_i|\theta)) \pi(\theta)}{\int_{\mathcal{T}} (\prod_{i=1}^n p(x_i|\theta')) \pi(\theta')d\theta'} \quad (8)$$

This can be written proportionally as:

$$\pi(\theta|\mathcal{D}) \propto \left(\prod_{i=1}^n p(x_i|\theta) \right) \pi(\theta)$$

Example 5.3 (Unknown Gaussian Mean). **Model.** Observations are generated by a Gaussian (Definition 2.1) with unknown mean θ and fixed, known standard deviation σ .

$$M = \{p(\cdot|\theta, \sigma) \mid \theta \in \mathbb{R}\}, \quad p(x|\theta, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\theta)^2}{2\sigma^2}\right).$$

Prior. We choose a Gaussian prior $\pi(\theta|\mu, \xi)$ with mean μ and standard deviation ξ .

Posterior. Applying Bayes' rule (Theorem 2.4) via Proposition 5.2, the posterior is again Gaussian:

$$\pi(\theta|x_{1:n}, \mu, \xi) = \pi(\theta|\mu_n, \xi_n),$$

where

$$\mu_n = \frac{\sigma^2\mu + \xi^2 \sum_{i=1}^n x_i}{\sigma^2 + n\xi^2}, \quad \xi_n^2 = \frac{\sigma^2\xi^2}{\sigma^2 + n\xi^2}. \quad (9)$$

Example 5.4 (Bayesian Coin Toss). **Model.** Coin toss with parameter $q \in [0, 1]$:

$$p(H|q) = q, \quad p(T|q) = 1 - q.$$

Prior. Choose a Beta distribution as prior:

$$p(q|\alpha_1, \alpha_2) = \frac{1}{B(\alpha_1, \alpha_2)} q^{\alpha_1-1} (1-q)^{\alpha_2-1}.$$

Posterior after one observation. If we observe a single head (H), by Proposition 5.2:

$$p(q|H) \propto p(q) p(H|q) \propto q \cdot q^{\alpha_1-1} (1-q)^{\alpha_2-1} = q^{\alpha_1} (1-q)^{\alpha_2-1}.$$

Thus the posterior is

$$p(q|H) = \text{Beta}(q|\alpha_1 + 1, \alpha_2).$$

Posterior after multiple observations. Suppose $D = \{HHTHTT\}$, with $N_H = 3$ heads and $N_T = 3$ tails. Then

$$p(D|q) = q^{N_H}(1-q)^{N_T} = q^3(1-q)^3.$$

The posterior is

$$p(q|D) \propto p(q)p(D|q) = q^{\alpha_1-1}(1-q)^{\alpha_2-1} \cdot q^3(1-q)^3 = q^{\alpha_1+2}(1-q)^{\alpha_2+2}.$$

Thus

$$p(q|D) = \text{Beta}(q|\alpha_1 + 3, \alpha_2 + 3).$$

5.2 Bayesian Model Selection

Bayesian inference (Definition 4.2) can be extended to compare different models.

Definition 5.5 (Marginal Likelihood / Evidence). Given a model $\mathcal{M}_i$ with parameters θ_i and prior $P(\theta_i|\mathcal{M}_i)$, the **marginal likelihood** (or **evidence**) for model $\mathcal{M}_i$ is the probability of the data integrated over all possible parameter values:

$$P(\mathcal{D}|\mathcal{M}_i) = \int P(\mathcal{D}|\theta_i, \mathcal{M}_i)P(\theta_i|\mathcal{M}_i)d\theta_i \quad (10)$$

The evidence for a model represents how well that model explains the data on average, according to its prior. We can use Bayes' rule (Theorem 2.4) at the level of models to compute the posterior probability of a model given the data:

$$P(\mathcal{M}_i|\mathcal{D}) = \frac{P(\mathcal{D}|\mathcal{M}_i)P(\mathcal{M}_i)}{P(\mathcal{D})} \propto P(\mathcal{D}|\mathcal{M}_i)P(\mathcal{M}_i) \quad (11)$$

where $P(\mathcal{M}_i)$ is a prior probability over the models themselves. This allows for principled comparison between different model structures.

Posterior Representation

In Bayesian inference (Definition 4.2) we often summarize as:

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}.$$

This captures the update from prior belief to posterior belief after observing data.

5.3 Conjugate Priors

In many cases, the integral in the denominator of the posterior (the evidence, Definition 5.5) is intractable. However, for certain choices of prior and likelihood, the calculation becomes simple.

Definition 5.6 (Conjugate Prior). A family of prior distributions Π is said to be **conjugate** to a family of likelihood functions $\{p(x|\theta)\}$ if for every prior $\pi \in \Pi$, the corresponding posterior distribution $\pi(\theta|x)$ is also in Π .

Theorem 5.7 (Conjugacy for Exponential Families). *Let the likelihood $p(x|\theta)$ be a member of the exponential family (Definition 3.1):*

$$p(x|\theta) = f(x)g(\theta)e^{\phi(\theta)^T T(x)}$$

A conjugate prior (Definition 5.6) for this family has the form:

$$p(\theta) = F(\tau, \nu)g(\theta)^\nu e^{\phi(\theta)^T \tau} \quad (12)$$

where τ and ν are hyperparameters of the prior. Given n conditionally i.i.d. observations $\{x_i\}$, the posterior distribution is of the same form:

$$p(\theta|\{x_i\}) \propto g(\theta)^{\nu+n} e^{\phi(\theta)^T(\tau + \sum_i T(x_i))} \quad (13)$$

The hyperparameters are updated as $\nu \rightarrow \nu + n$ and $\tau \rightarrow \tau + \sum_i T(x_i)$.

Remark 5.8 (Interpretation of Conjugate Updates). • The hyperparameter ν can be interpreted as a count of "pseudo-observations" that express the strength of our prior belief. The posterior strength is $\nu + n$.

- The hyperparameter τ can be interpreted as the sufficient statistics of these pseudo-observations. The posterior sufficient statistic is the sum of the prior's and the data's sufficient statistics.
- The posterior effectively interpolates between the prior belief (represented by τ) and the observational evidence (represented by $\sum_i T(x_i)$).

Example 5.9 (Conjugacy in the Coin Flip). The Bernoulli distribution for a single coin flip ($x \in \{0, 1\}$) with parameter q can be written in exponential family form (Definition 3.1):

$$P(x|q) = q^x(1-q)^{1-x} = (1-q)e^{x \log(q/(1-q))}$$

Here the natural parameter is the log-odds $\log(q/(1-q))$ and the sufficient statistic is $T(x) = x$. The conjugate prior (Definition 5.6) is the Beta distribution:

$$\text{Beta}(q|\alpha_1, \alpha_2) = \frac{1}{B(\alpha_1, \alpha_2)} q^{\alpha_1-1} (1-q)^{\alpha_2-1}$$

If the prior is $\text{Beta}(q|\alpha_1, \alpha_2)$ and we observe data $\mathcal{D}$ with N_H heads and N_T tails, the posterior is $\text{Beta}(q|\alpha_1 + N_H, \alpha_2 + N_T)$, confirming Theorem 5.7.

6 Maximum Likelihood Estimation in Detail

Definition 6.1 (Maximum Likelihood Estimator). Given a dataset $\mathcal{D} = \{x_1, \dots, x_n\}$ generated i.i.d. (Definition 2.7) from a model $\mathcal{M} = \{p(x|\theta) \mid \theta \in \mathcal{T}\}$, the **Maximum Likelihood Estimator (MLE)** is defined as

$$\hat{\theta}_{\text{ML}} := \arg \max_{\theta \in \mathcal{T}} p(x_1, \dots, x_n|\theta) = \arg \max_{\theta \in \mathcal{T}} \prod_{i=1}^n p(x_i|\theta) \quad (14)$$

Proposition 6.2 (Log-Likelihood Trick). *Since the logarithm is a monotonically increasing function, maximizing a function is equivalent to maximizing its logarithm. It is almost always easier to work with the **log-likelihood**, $\ell(\theta) = \log p(\mathcal{D}|\theta)$.*

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta} \log \left(\prod_{i=1}^n p(x_i|\theta) \right) = \arg \max_{\theta} \sum_{i=1}^n \log p(x_i|\theta) \quad (15)$$

If the log-likelihood function is differentiable, the MLE (Definition 6.1) can often be found by solving the **maximum likelihood equation**:

$$\nabla_{\theta} \ell(\theta) = \sum_{i=1}^n \nabla_{\theta} \log p(x_i|\theta) = 0 \quad (16)$$

Theorem 6.3 (MLE and the Law of Large Numbers). *The MLE (Definition 6.1) can be derived from the Law of Large Numbers (LLN). The LLN states that for i.i.d. random variables X_i and a function f ,*

$$\frac{1}{n} \sum_{i=1}^n f(X_i) \xrightarrow{n \rightarrow \infty} \mathbb{E}[f(X_1)] \quad \text{with probability 1.}$$

*Suppose the true data-generating parameter is θ_0 . Consider the function $g(x, \theta) := \nabla_{\theta} \log p(x, \theta)$, known as the **score function**. Under suitable regularity conditions, it can be shown that*

$$\mathbb{E}_{P_{\theta_0}}[g(X, \theta)] = \mathbb{E}_{P_{\theta_0}}[\nabla_{\theta} \log p(X, \theta)] = 0 \quad \text{if and only if } \theta = \theta_0. \quad (17)$$

By the LLN, we can estimate this expectation with the sample mean:

$$\frac{1}{n} \sum_{i=1}^n g(X_i, \theta) = \frac{1}{n} \sum_{i=1}^n \nabla_{\theta} \log p(X_i, \theta) \approx 0$$

The solution to this equation for θ is exactly the maximum likelihood estimator, recovering the ML equation (16).

7 Review: Gaussian Distributions and Linear Algebra

7.1 Gaussian Distributions

Definition 7.1 (Multivariate Gaussian Distribution). A random vector $\mathbf{x} \in \mathbb{R}^D$ follows a **multivariate Gaussian (or Normal) distribution**, denoted $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, if its probability density function (PDF) is given by:

$$p(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\det(\boldsymbol{\Sigma})|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right) \quad (18)$$

where $\boldsymbol{\mu} \in \mathbb{R}^D$ is the mean vector and $\boldsymbol{\Sigma} \in \mathbb{R}^{D \times D}$ is the symmetric, positive definite **covariance matrix**.

Proposition 7.2 (Properties of the Covariance Matrix). *The covariance matrix $\boldsymbol{\Sigma}$ in Definition 7.1 captures the dependencies between the dimensions of the random vector $\mathbf{x}$.*

- The diagonal elements $\Sigma_{ii} = \text{Var}(X_i)$ are the variances of the individual components.
- The off-diagonal elements $\Sigma_{ij} = \text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)]$ are the covariances between components.
- Since $\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$, the matrix $\boldsymbol{\Sigma}$ is symmetric.
- The quadratic form $\mathbf{z}^T \boldsymbol{\Sigma} \mathbf{z}$ represents the variance of the random variable $\mathbf{z}^T \mathbf{x}$, which must be non-negative. This implies $\boldsymbol{\Sigma}$ is positive semi-definite. For the density to be well-defined, it must be invertible, hence positive definite.

7.2 Eigen-decomposition and Geometry of Gaussians

The geometry of the multivariate Gaussian (Definition 7.1) is determined by the spectral properties of its covariance matrix $\boldsymbol{\Sigma}$.

Theorem 7.3 (Spectral Theorem for Symmetric Matrices). *If a matrix $\mathbf{A} \in \mathbb{R}^{D \times D}$ is symmetric, then:*

1. All its eigenvalues are real numbers.

2. It has D eigenvectors that form an orthonormal basis for $\mathbb{R}^D$.
3. It can be diagonalized as $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$, where $\mathbf{U}$ is an orthogonal matrix whose columns are the eigenvectors of $\mathbf{A}$, and $\mathbf{\Lambda}$ is a diagonal matrix of the corresponding eigenvalues.

Proposition 7.4 (Geometry of a Gaussian). *Let $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ (Definition 7.1).*

1. *The surfaces of constant probability density are ellipsoids defined by the equation $(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) = c$ for some constant $c > 0$. This term is known as the Mahalanobis distance.*
2. *The principal axes of these ellipsoids are aligned with the eigenvectors of $\boldsymbol{\Sigma}$ (by the spectral decomposition in Theorem 7.3).*
3. *The lengths of these principal axes are proportional to the square roots of the corresponding eigenvalues of $\boldsymbol{\Sigma}$.*
4. *A multidimensional Gaussian can be understood as a set of independent, 1D Gaussian variables in the coordinate system defined by the eigenvectors of $\boldsymbol{\Sigma}$. In this rotated and shifted coordinate system, the covariance matrix is diagonal, with the eigenvalues of $\boldsymbol{\Sigma}$ on the diagonal.*

8 Applications of Estimation

8.1 MLE for a Gaussian Distribution

Theorem 8.1 (MLE for Gaussian Parameters). *Given an i.i.d. dataset $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ from a Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ (Definition 7.1), the maximum likelihood estimators (Definition 6.1) for the mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$ are the sample mean and the sample covariance:*

$$\hat{\boldsymbol{\mu}}_{ML} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \quad (19)$$

$$\hat{\boldsymbol{\Sigma}}_{ML} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_{ML})(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_{ML})^T \quad (20)$$

Proof. The log-likelihood (Proposition 6.2) for n data points is:

$$\ell(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{i=1}^n \log p(\mathbf{x}_i | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = -\frac{nD}{2} \log(2\pi) - \frac{n}{2} \log |\det(\boldsymbol{\Sigma})| - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})$$

To find the MLE for $\boldsymbol{\mu}$, we take the gradient with respect to $\boldsymbol{\mu}$ and set it to zero:

$$\begin{aligned} \nabla_{\boldsymbol{\mu}} \ell &= -\frac{1}{2} \sum_{i=1}^n \nabla_{\boldsymbol{\mu}} ((\mathbf{x}_i - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})) \\ &= -\frac{1}{2} \sum_{i=1}^n 2\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \mathbf{x}_i) = \boldsymbol{\Sigma}^{-1} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu}) = 0 \end{aligned}$$

Since $\boldsymbol{\Sigma}^{-1}$ is invertible, this implies $\sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu}) = 0$, which gives $\hat{\boldsymbol{\mu}}_{ML} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$.

For $\hat{\boldsymbol{\Sigma}}_{ML}$, we take the derivative with respect to $\boldsymbol{\Sigma}^{-1}$ (this is easier) and set it to zero, using matrix derivative identities $\frac{\partial}{\partial \mathbf{A}} \log |\mathbf{A}^{-1}| = -\mathbf{A}^T$ and $\frac{\partial}{\partial \mathbf{A}} \text{Tr}(\mathbf{BAC}) = \mathbf{B}^T \mathbf{C}^T$.

$$\nabla_{\boldsymbol{\Sigma}^{-1}} \ell = \frac{n}{2} \boldsymbol{\Sigma}^T - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T = 0$$

Since $\boldsymbol{\Sigma}$ is symmetric, $\boldsymbol{\Sigma}^T = \boldsymbol{\Sigma}$. Solving for $\boldsymbol{\Sigma}$ gives the result. $\square$

8.2 Multivariate Linear Regression

A simple form of supervised learning (Definition 1.1) where we model the output $\mathbf{y}$ as a linear function of the input $\mathbf{x}$, corrupted by Gaussian noise.

Definition 8.2 (Multivariate Linear Regression Model). The conditional distribution of $\mathbf{y} \in \mathbb{R}^{D_y}$ given $\mathbf{x} \in \mathbb{R}^{D_x}$ is modeled as:

$$p(\mathbf{y}|\mathbf{x}, \mathbf{W}, \Sigma_y) = \mathcal{N}(\mathbf{y}|\mathbf{W}\mathbf{x}, \Sigma_y) \quad (21)$$

where $\mathbf{W}$ is a $D_y \times D_x$ weight matrix and Σ_y is the covariance of the noise.

8.2.1 ML Estimation for Linear Regression

Given a dataset $\mathcal{D} = \{(\mathbf{x}_1, \mathbf{y}_1), \dots, (\mathbf{x}_N, \mathbf{y}_N)\}$, the conditional log-likelihood (Proposition 6.2) is:

$$\ell(\mathbf{W}) = \sum_{i=1}^N \log p(\mathbf{y}_i|\mathbf{x}_i, \mathbf{W}, \Sigma_y) = -\frac{1}{2} \sum_{i=1}^N (\mathbf{y}_i - \mathbf{W}\mathbf{x}_i)^T \Sigma_y^{-1} (\mathbf{y}_i - \mathbf{W}\mathbf{x}_i) + \text{const.}$$

Maximizing this is equivalent to minimizing the sum of squared errors term. Taking the derivative with respect to $\mathbf{W}$ and setting to zero yields the solution:

$$\hat{\mathbf{W}}_{\text{ML}} = \left(\sum_{i=1}^N \mathbf{y}_i \mathbf{x}_i^T \right) \left(\sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^T \right)^{-1} \quad (22)$$

This is the well-known normal equations solution.

8.2.2 Bayesian Linear Regression

Let's consider a scalar output y_i and a weight vector $\mathbf{w}$. The model (Definition 8.2) is $y_i \sim \mathcal{N}(\mathbf{w}^T \mathbf{x}_i, \sigma_y^2)$. We can place a Gaussian prior on the weights:

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1})$$

The log posterior on $\mathbf{w}$ is (by Theorem 2.4):

$$\begin{aligned} \log p(\mathbf{w}|\mathcal{D}) &= \log p(\mathcal{D}|\mathbf{w}) + \log p(\mathbf{w}) - \log p(\mathcal{D}) \\ &\propto -\frac{1}{2\sigma_y^2} \sum_i (y_i - \mathbf{w}^T \mathbf{x}_i)^2 - \frac{1}{2} \mathbf{w}^T \mathbf{A} \mathbf{w} \end{aligned}$$

This is a quadratic form in $\mathbf{w}$, so the posterior $p(\mathbf{w}|\mathcal{D})$ is also Gaussian.

Proposition 8.3 (MAP Estimate and Ridge Regression). *The MAP estimate (Definition 4.4) for $\mathbf{w}$ is the mode of the Gaussian posterior. For a scalar output and prior $p(\mathbf{w}) = \mathcal{N}(\mathbf{0}, \alpha^{-1}\mathbf{I})$, finding the MAP estimate is equivalent to minimizing:*

$$\sum_i (y_i - \mathbf{w}^T \mathbf{x}_i)^2 + \alpha \|\mathbf{w}\|^2 \quad (23)$$

*This is the objective function for **Ridge Regression**. The prior acts as a regularizer, shrinking the weights towards the prior mean (zero), which helps prevent overfitting.*